

```
1 1
2 1
3 1
4 1

dtype: int64

[50] ✓ 0s print("X shape:", x.shape)
print("y shape:", y.shape)

X shape: (980, 7)
y shape: (980,)

[51] ✓ 0s clean_df.to_csv('cleaned_loan_approval.csv', index=False)

[52] ✓ 0s files.download('cleaned_loan_approval.csv')

[ ] Start coding or generate with AI.
```

Variables Terminal

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	age	income_lakh	credit_score	loan_amount_lakh	existing_loans	approved	debt_burden	loan_to_income
0	28.0	12.64	613.0	8.93	3.0	0.237342	0.706487	
1	57.0	19.80	753.0	28.22	3.0	0.151515	1.425253	

```
[46] X = clean_df.drop('approved', axis=1)
[47] y = clean_df['approved']
[48] X.head()
```

	age	income_lakh	credit_score	loan_amount_lakh	existing_loans	debt_burden	loan_to_income
0	28.0	12.64	613.0	8.93	3.0	0.237342	0.706487
1	57.0	19.80	753.0	28.22	3.0	0.151515	1.425253

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0	28.0	12.64	613.0	8.93	3.0	1	0.237342	0.706
1	57.0	19.80	753.0	28.22	3.0	1	0.151515	1.425
2	35.0	20.54	787.0	18.94	1.0	1	0.048685	0.922
3	59.0	18.87	816.0	11.61	1.0	1	0.052994	0.615
4	62.0	11.63	553.0	5.23	3.0	1	0.257954	0.449

Next steps: [Generate code with clean_df](#) [New interactive sheet](#)

```
[43] clean_df.shape
✓ Os (980, 8)
```

```
[44] clean_df.isnull().sum()
✓ Os
```

	0
age	0
income_lakh	0
credit_score	0
loan_amount_lakh	0
existing_loans	0

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Back up now

50%	42.000000	13.765000	643.000000	15.845000	2.000000	1.000000
75%	53.000000	19.405000	743.000000	22.647500	3.000000	1.000000
max	64.000000	24.990000	850.000000	29.960000	4.000000	1.000000

```
[38] clean_df['debt_burden'] = (  
    clean_df['existing_loans'] /  
    clean_df['income_lakh'].replace(0, np.nan)  
)
```

```
[39] clean_df[['existing_loans', 'income_lakh', 'debt_burden']].head()
```

	existing_loans	income_lakh	debt_burden
0	3.0	12.64	0.237342
1	3.0	19.80	0.151515
2	1.0	20.54	0.048685
3	1.0	18.87	0.052994
4	3.0	11.63	0.257954

```
[40] clean_df['loan_to_income'] = (  
    clean_df['loan_amount_lakh'] /  
    clean_df['income_lakh'].replace(0, np.nan)  
)
```

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```
[35] ✓ Os clean_df = df.copy()

[36] ✓ Os clean_df = clean_df.drop_duplicates()

clean_df = clean_df[
    (clean_df['age'] >= 18) &
    (clean_df['age'] <= 100)
]

clean_df = clean_df[
    (clean_df['credit_score'] >= 300) &
    (clean_df['credit_score'] <= 900)
]

clean_df = clean_df[
    clean_df['income_lakh'] >= 0
]

clean_df = clean_df[
    clean_df['existing_loans'] >= 0
]

[37] ✓ Os clean_df.describe()
```

	age	income_lakh	credit_score	loan_amount_lakh	existing_loans	approved
count	980.000000	980.000000	980.000000	980.000000	980.000000	980.000000
mean	42.502041	13.518041	642.466327	15.755714	2.061224	0.662245

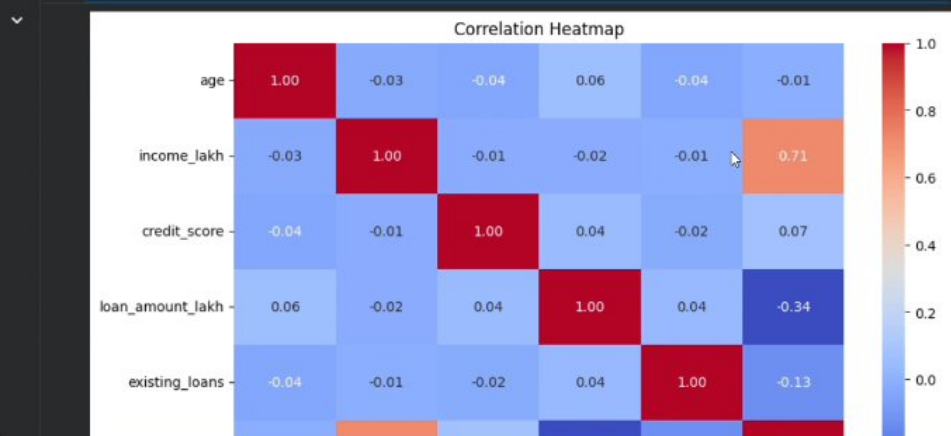
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```
[34]  
✓ Os  
correlation = df.corr(numeric_only=True)  
  
sns.heatmap(correlation, annot=True, cmap='coolwarm', fmt='.2f')  
  
plt.title('Correlation Heatmap')  
  
plt.show()
```

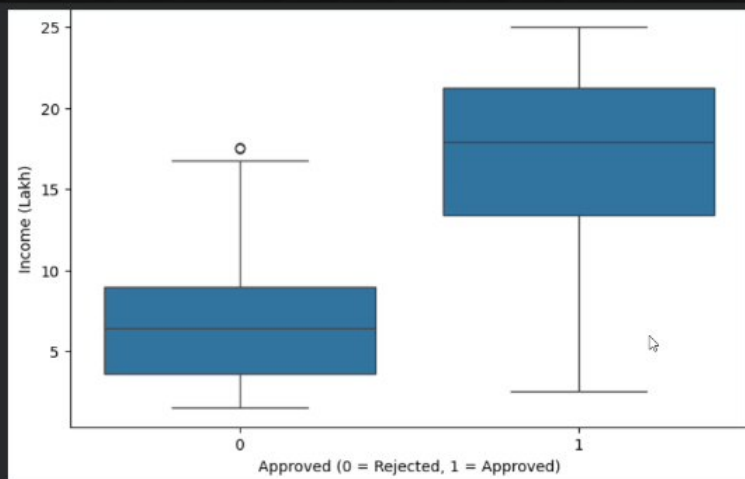


Variables Terminal

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```
[33]  
✓ 0s plt.figure(figsize=(8,5))  
  
sns.boxplot(x='approved', y='loan_amount_lakh', data=df)  
  
plt.title('Loan Amount vs Loan Approval')
```

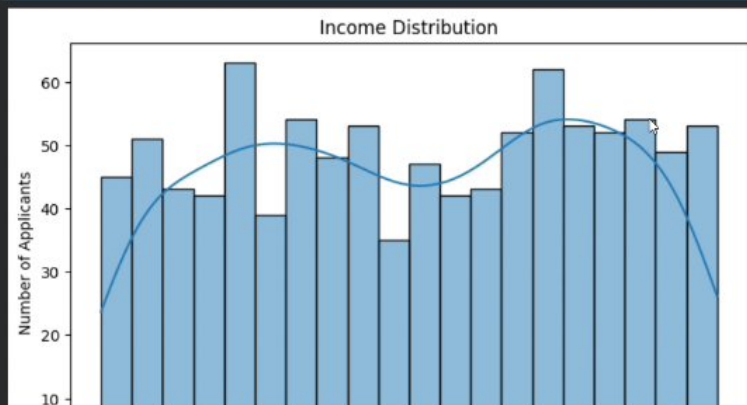
Variables Terminal

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```
[30]  
✓ Os  
plt.figure(figsize=(8,5))  
  
sns.histplot(df['income_lakh'], bins=20, kde=True)  
  
plt.title('Income Distribution')  
plt.xlabel('Income (Lakh)')  
plt.ylabel('Number of Applicants')  
  
plt.show()
```

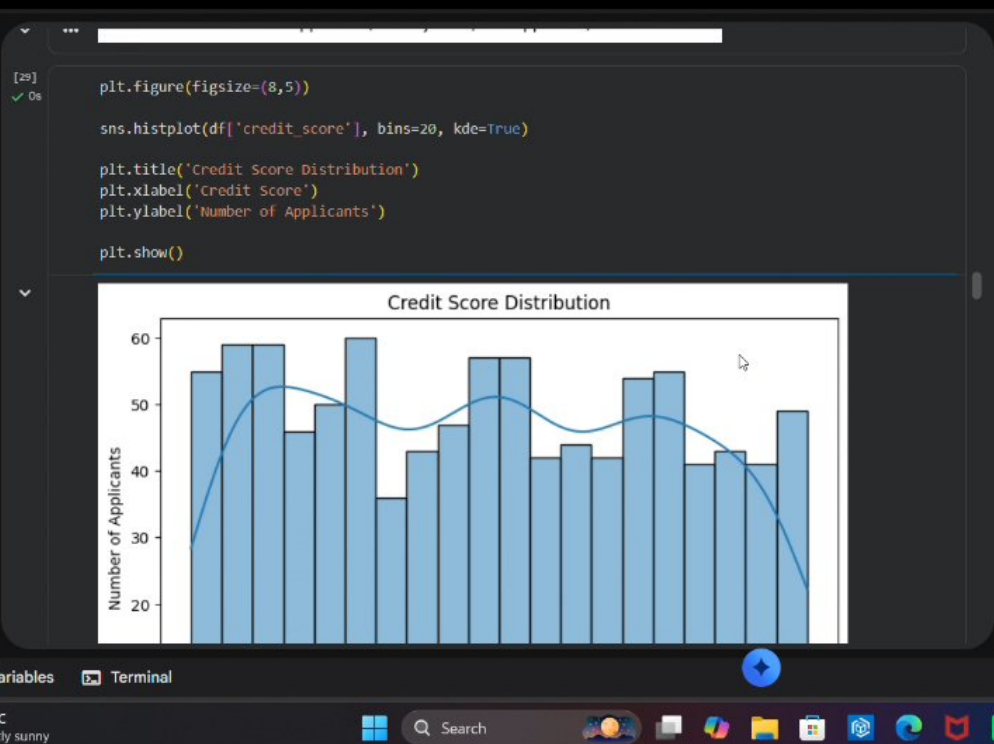


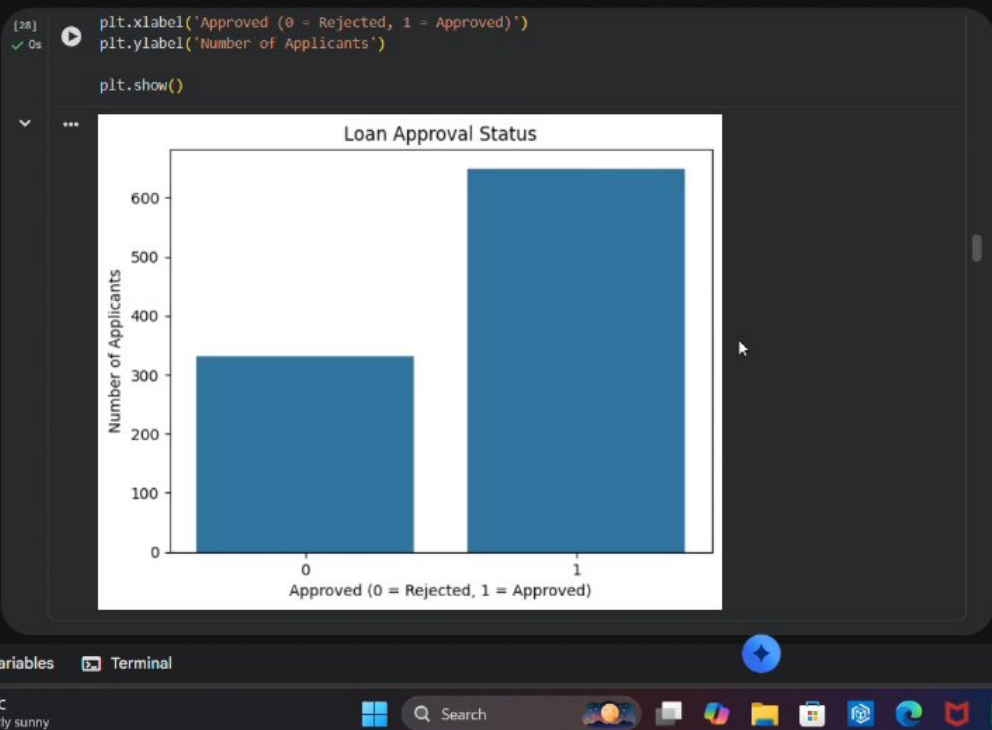
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[23] ✓ 0s

```
df[df['income_lakh'] < 0]
```

	age	income_lakh	credit_score	loan_amount_lakh	existing_loans	approved
20	39.0	-5.0	584.0	21.52	2.0	1
210	60.0	-5.0	476.0	17.56	1.0	1
565	47.0	-5.0	476.0	29.76	1.0	0
739	62.0	-5.0	687.0	6.70	1.0	1
812	31.0	-5.0	822.0	9.01	1.0	1

[24] ✓ 0s

```
df = df[df['income_lakh'] >= 0]
```

[25] ✓ 0s

```
df[df['existing_loans'] < 0]
```

	age	income_lakh	credit_score	loan_amount_lakh	existing_loans	approved
344	34.0	17.10	611.0	12.59	-2.0	1
500	36.0	19.08	490.0	25.68	-2.0	1
573	43.0	11.64	730.0	8.50	-2.0	1
821	35.0	17.91	538.0	5.98	-2.0	1
885	25.0	17.39	746.0	27.63	-2.0	1

Variables Terminal

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	age	income_lakh	credit_score	loan_amount_lakh	existing_loans	approved
20	39.0	-5.0	584.0	21.52	2.0	1



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Back up now

```
[16] ✓ Os df = df.drop_duplicates()
[17] ✓ Os df.duplicated().sum()
      np.int64(0)
[18] ✓ Os df['age'].describe()
      count    1000.000000
      mean     42.354000
      std      12.685725
      min      15.000000
      25%      31.000000
      50%      42.000000
      75%      53.000000
      max      64.000000
      dtype: float64
[19] ✓ Os df[(df['age'] < 18) | (df['age'] > 100)]
```

Variables Terminal

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Back up now

```
df.isnull().sum()

age      0
income_lakh  0
credit_score  0
loan_amount_lakh  0
existing_loans  0
approved  0

dtype: int64

[15] df.duplicated().sum()
np.int64(20)

[16] df = df.drop_duplicates()

[17] df.duplicated().sum()
np.int64(0)

[18] df['age'].describe()
```

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Back up now

```
[11] df.isnull().sum()
0
age      11
income_lakh  10
credit_score  10
loan_amount_lakh  10
existing_loans  10
approved  0
dtype: int64

[12] (df.isnull().sum() / len(df)) * 100
0
age      1.078431
income_lakh  0.980392
credit_score  0.980392
loan amount lakh  0.980392
```

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```
credit_score    float64
loan_amount_lakh float64
existing_loans   float64
approved        int64

dtype: object

[10]
✓ Os
df.describe()

   age  income_lakh  credit_score  loan_amount_lakh  existing_loans  approved
count  1009.000000   1010.000000   1010.000000      1010.000000      1010.000000  1020.000000
mean    42.448959    13.453802    644.527723      15.857168        2.048515    0.664706
std     12.753447     6.949768    119.393453      8.273434        1.423512    0.472325
min     15.000000    -5.000000   450.000000      1.040000       -2.000000    0.000000
25%     31.000000     7.372500   539.000000      8.995000        1.000000    0.000000
50%     42.000000    13.765000   643.000000     15.860000        2.000000    1.000000
75%     53.000000    19.387500   744.750000     22.900000        3.000000    1.000000
max     64.000000    24.990000  1000.000000     29.960000        4.000000    1.000000

[11]
✓ Os
df.isnull().sum()
```